

Project Proposal

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Project Title: A Multilevel and Robust Investigation of Delivery Lead Time in E-Commerce

1. Motivation

Delivery performance represents a key dimension of service quality in e-commerce platforms. Substantial heterogeneity in delivery lead times may arise from seller behavior, geographic distance, and order characteristics. Understanding the sources of such variability is relevant for performance monitoring and operational optimization. This project aims to investigate the determinants and hierarchical structure of delivery lead time using advanced multivariate methods.

2. Research Questions

The analysis will address the following research questions:

1. How much of the variability in delivery lead time is attributable to seller-level and geographic-level heterogeneity?
2. Which order and shipment characteristics are significantly associated with faster or slower delivery times?
3. Are the estimated effects robust to extreme observations and sampling variability?

3. Data

The project will use the [Olist Brazilian e-commerce dataset](#). The unit of analysis is the individual order. The primary outcome variable is delivery lead time, defined as the difference between the order purchase timestamp and the customer delivery date.

Explanatory variables will include selected order-level characteristics (e.g., freight value, product weight, payment characteristics), as well as geographic indicators (e.g., seller and customer state). The dataset will be cleaned, restricted to completed deliveries, and examined for missing values and extreme observations.

4. Methodology

The methodological framework follows the Advanced Multivariate Statistics syllabus.

- **Exploratory Data Analysis:** Descriptive statistics and visualization of delivery time distribution and heterogeneity across sellers and regions.

- **Baseline Regression Model:** A linear model (possibly using log-transformed delivery time) to assess associations between delivery time and order characteristics.
- **Multilevel Modeling:** A mixed-effects model including random intercepts for sellers and/or geographic units (e.g., customer state) to quantify hierarchical variance components and intra-class correlation.
- **Robustness Analysis:** Sensitivity analyses excluding extreme delivery times and bootstrap procedures to evaluate the stability of key parameter estimates.

5. Expected Contribution

The project aims to quantify the relative importance of seller-level and geographic heterogeneity in delivery performance, identify significant operational drivers of delivery time, and demonstrate the relevance of multilevel and robust methods in applied e-commerce analytics.

6. Software

The analysis will be conducted in R using packages for mixed-effects modeling, robust inference, and bootstrap estimation.